

GLmomentEst

An R package for Generalized L-moments Estimation of the Generalized Extreme Value (GEV) Distribution.

Features

- **Stationary GEV estimation** (`glme.gev`): Estimate GEV parameters using generalized L-moments with various penalty functions
- **Non-stationary GEV11 estimation** (`glme.gev11`): Estimate time-varying GEV parameters where location and scale change linearly with time
- **Multiple penalty functions**: `beta`, `norm`, `ms` (Martins-Stedinger), `park`, `cannon`, `cd`, and `no` (no penalty)

Installation

Install the latest development version (on GitHub) via `{remotes}`:

```
remotes::install_github("sygstat/GL-momentEst")
```

Getting started

Stationary GEV estimation

```
library(GLmomentEst)

# Load example data
data("streamflow")

# Basic usage with default beta penalty
result <- glme.gev(streamflow$r1)
result$glme # GLME estimates (location, scale, shape)
result$lme  # L-moment estimates for comparison

# With different penalty functions
glme.gev(streamflow$r1, pen = "norm", mu = -0.5, std = 0.2)
glme.gev(streamflow$r1, pen = "ms")      # Martins-Stedinger
glme.gev(streamflow$r1, pen = "park")    # Park
glme.gev(streamflow$r1, pen = "cannon")  # Cannon
glme.gev(streamflow$r1, pen = "cd")      # CD
glme.gev(streamflow$r1, pen = "no")      # No penalty

# Using preset hyperparameter choices
glme.gev(streamflow$r1, pen = "beta", pen.choice = 2)
glme.gev(streamflow$r1, pen = "norm", pen.choice = 3)
```

Non-stationary GEV11 estimation

The GEV11 model assumes time-varying parameters:

- Location: $\mu(t) = \mu_0 + \mu_1 * t$
- Scale: $\sigma(t) = \exp(\sigma_0 + \sigma_1 * t)$
- Shape: ξ (constant)

```
library(GLmomentEst)

# Load example data
data("PhliuAgromet")

# Estimate non-stationary GEV11 model
result <- glme.gev11(PhliuAgromet$prec)

# Results
result$para.prop    # Proposed GLME estimates (mu0, mu1, sigma0, sigma1,
xi)
result$lme.sta      # Stationary L-moment estimates
result$para.gado    # GN16 original estimates
result$strup.final # Weighted least squares estimates
```

Function Reference

Main Functions

Function	Description
<code>glme.gev()</code>	Stationary GEV parameter estimation using GLME
<code>glme.gev11()</code>	Non-stationary GEV11 model estimation
<code>glme.like()</code>	Likelihood function for GLME optimization
<code>init.glme()</code>	Initialize parameters for optimization

Penalty Functions

Penalty	Description	Hyperparameters
<code>beta</code>	Adaptive beta distribution (default)	<code>p</code> , <code>c1</code> , <code>c2</code>
<code>norm</code>	Normal distribution	<code>mu</code> , <code>std</code>
<code>ms</code>	Martins-Stedinger (Beta(6,9))	-
<code>park</code>	Park (Beta(2.5,2.5))	-
<code>cannon</code>	Cannon (Beta(2,3,3))	-
<code>cd</code>	CD penalty	-

Penalty	Description	Hyperparameters
no	No penalty	-

Authors

- **Yonggwon Shin**, Ph.D, Research & Development Center, XRAI Inc., Gwangju 61186, Korea, syg.stat@gmail.com
- **Yire Shin**, Ph.D, Department of Mathematics and Statistics, Chonnam National University, Republic of Korea, shinyire87@gmail.com
- **Jeong-Soo Park**, Professor, Department of Mathematics and Statistics, Chonnam National University, Republic of Korea, jspark@jnu.ac.kr

Citation

Shin, Y., Shin, Y., Park, J. S. (2024). Generalized L-moments estimation, applied to the generalized extreme value distribution.

License

GPL (>= 3)
